

STAT293 - Simulation Result Plots

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Table 1 lists the 16 simulation settings, and Table 2 summarizes the common parameter values (with heterogeneity fixed at the medium level).

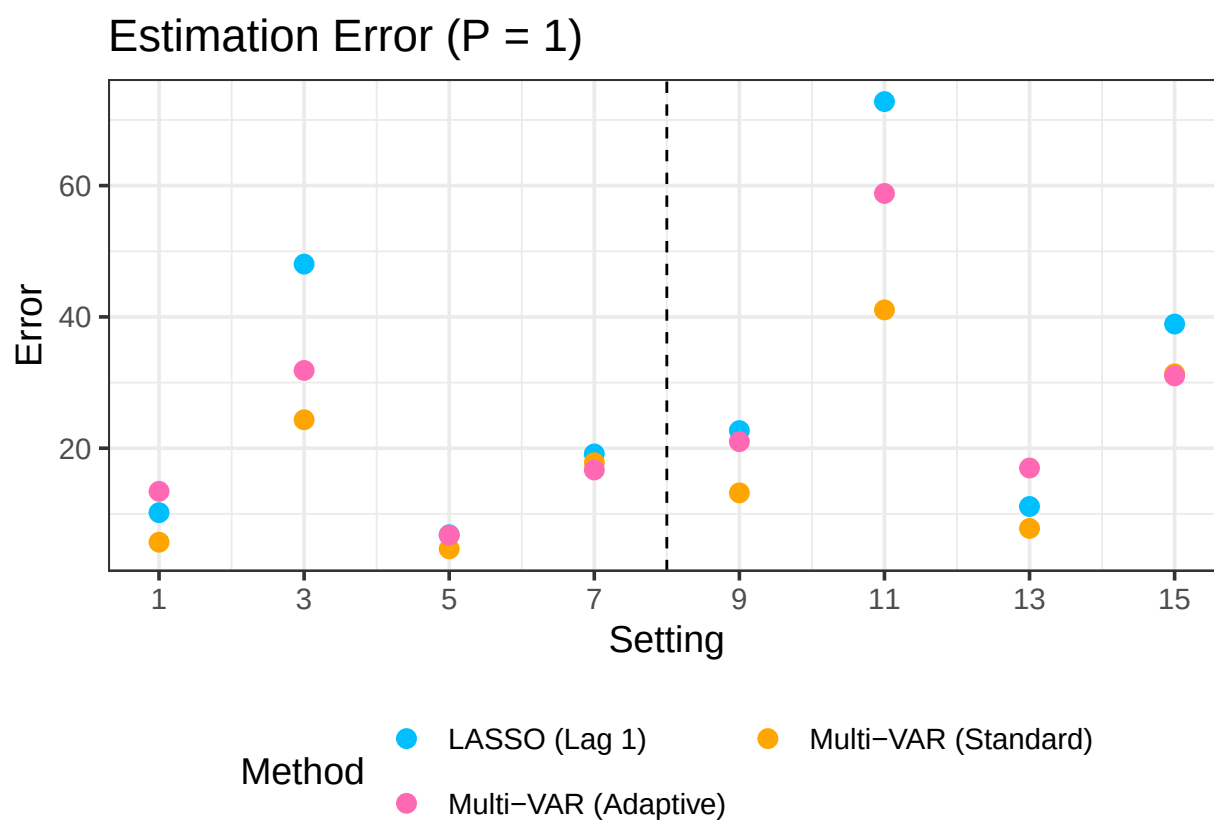
Table 1. Simulation settings: each row corresponds to one design configuration (indexed by “Setting No.”).

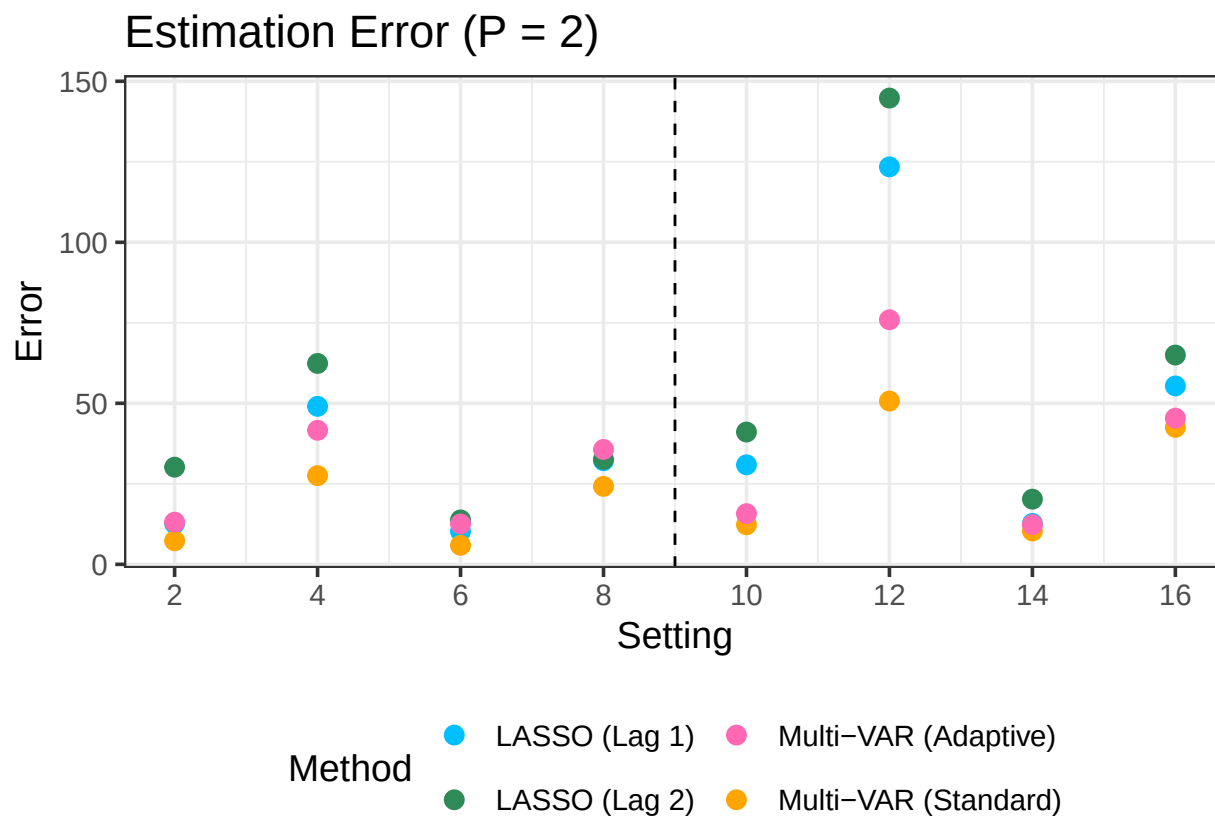
Setting No.	K	T	d	p
1	10	30	10	1
2	10	30	10	2
3	10	30	20	1
4	10	30	20	2
5	10	50	10	1
6	10	50	10	2
7	10	50	20	1
8	10	50	20	2
9	20	30	10	1
10	20	30	10	2
11	20	30	20	1
12	20	30	20	2
13	20	50	10	1
14	20	50	10	2
15	20	50	20	1
16	20	50	20	2

Table 2. Common Parameter Values Used in All Simulation Settings

Parameter	Value	Description
sparsity	0.05	Proportion of non-zero coefficients
heterogeneity	medium	Level of heterogeneity (“low”, “medium”, “high”)
sigma_eps	0.1	Standard deviation of noise
phi_max	0.9	Upper bound for coefficient generation
phi_min	0.1	Lower bound for coefficient generation
n_holdout	5	Number of time points withheld for forecasting

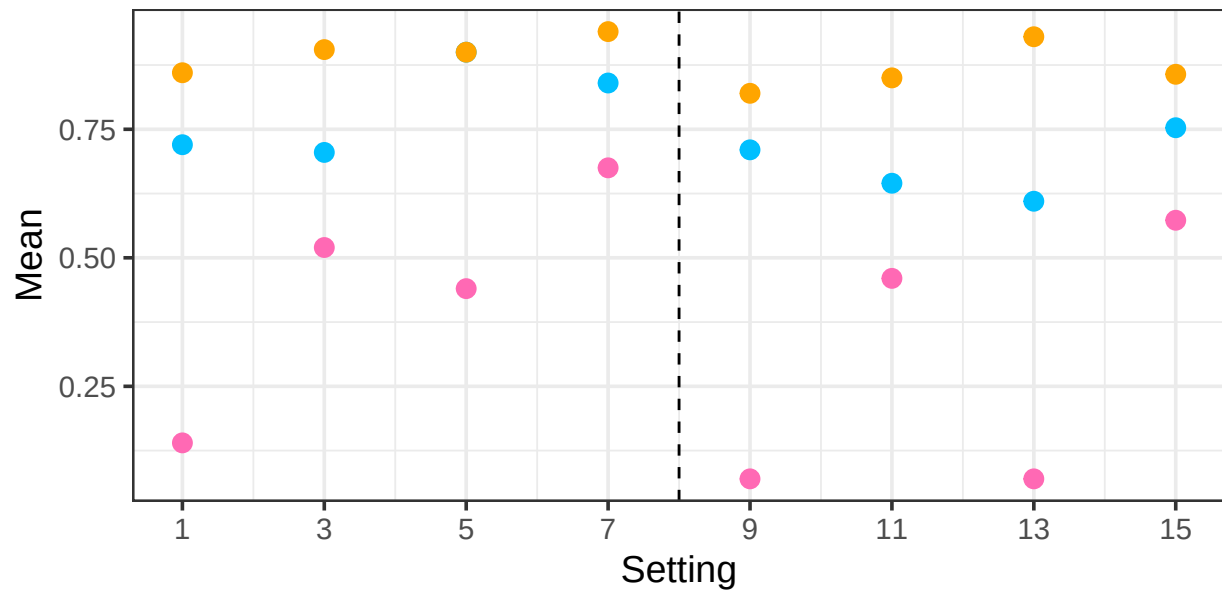
ESTIMATION ERROR





SENSITIVITY

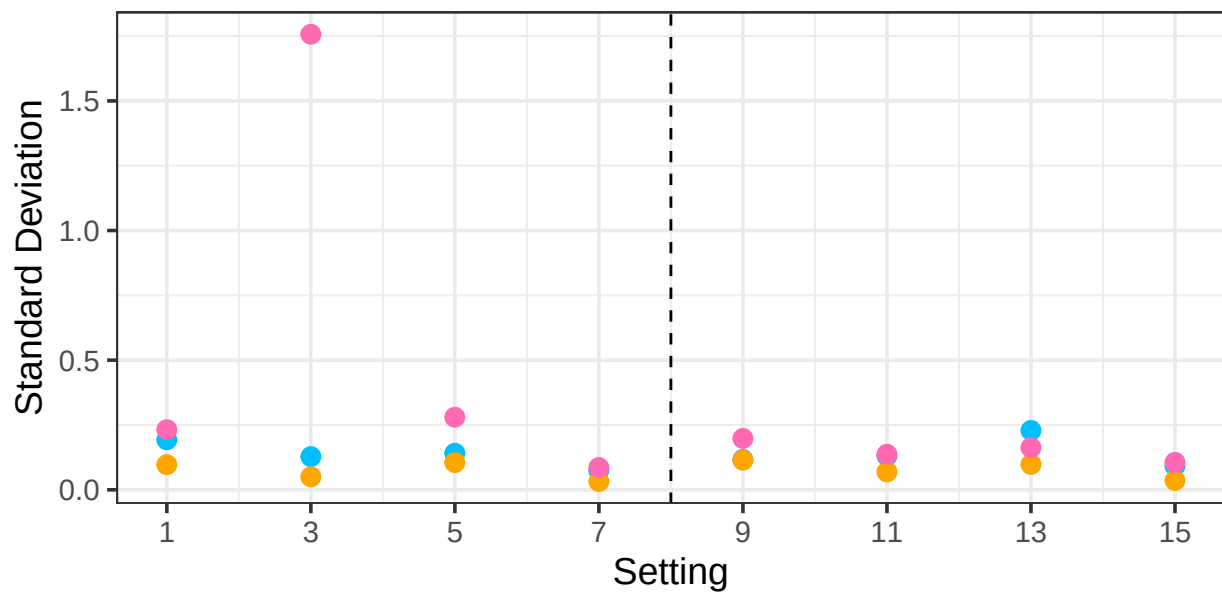
Means of Sensitivity ($P = 1$)



Method

- LASSO (Lag 1)
- Multi-VAR (Standard)
- Multi-VAR (Adaptive)

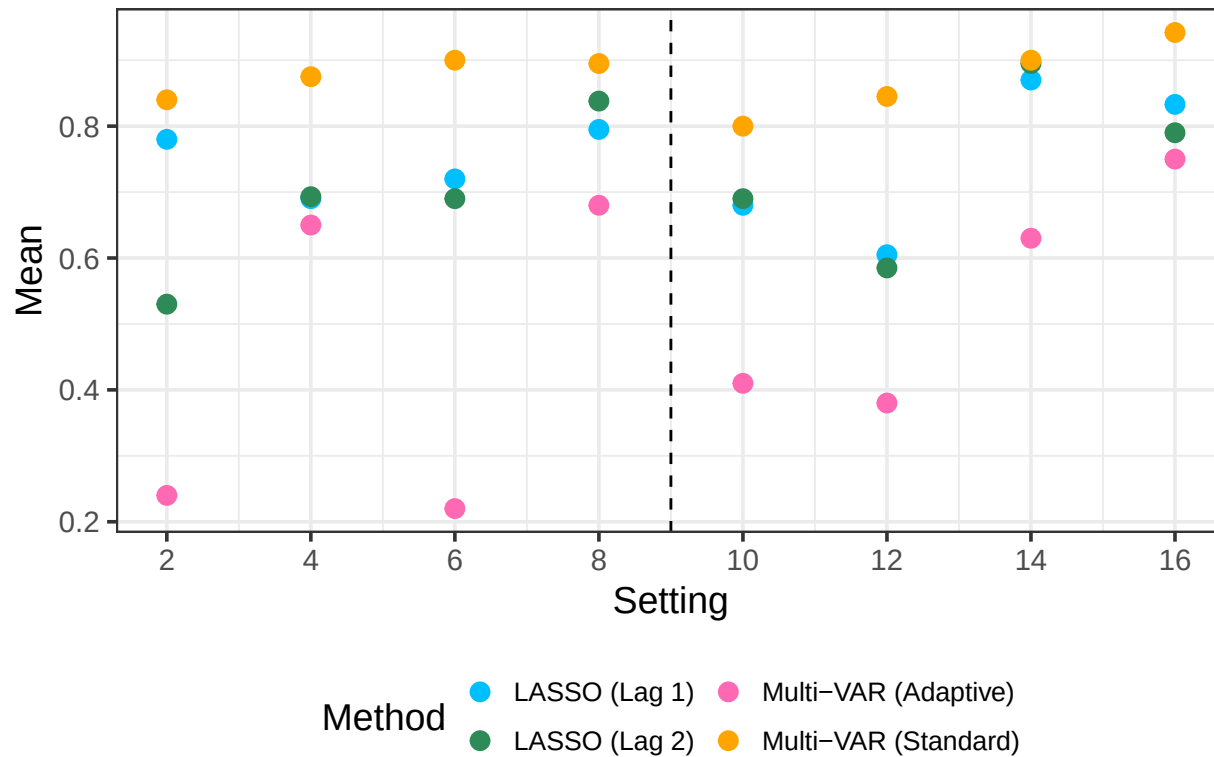
Standard Deviations of Sensitivity ($P = 1$)



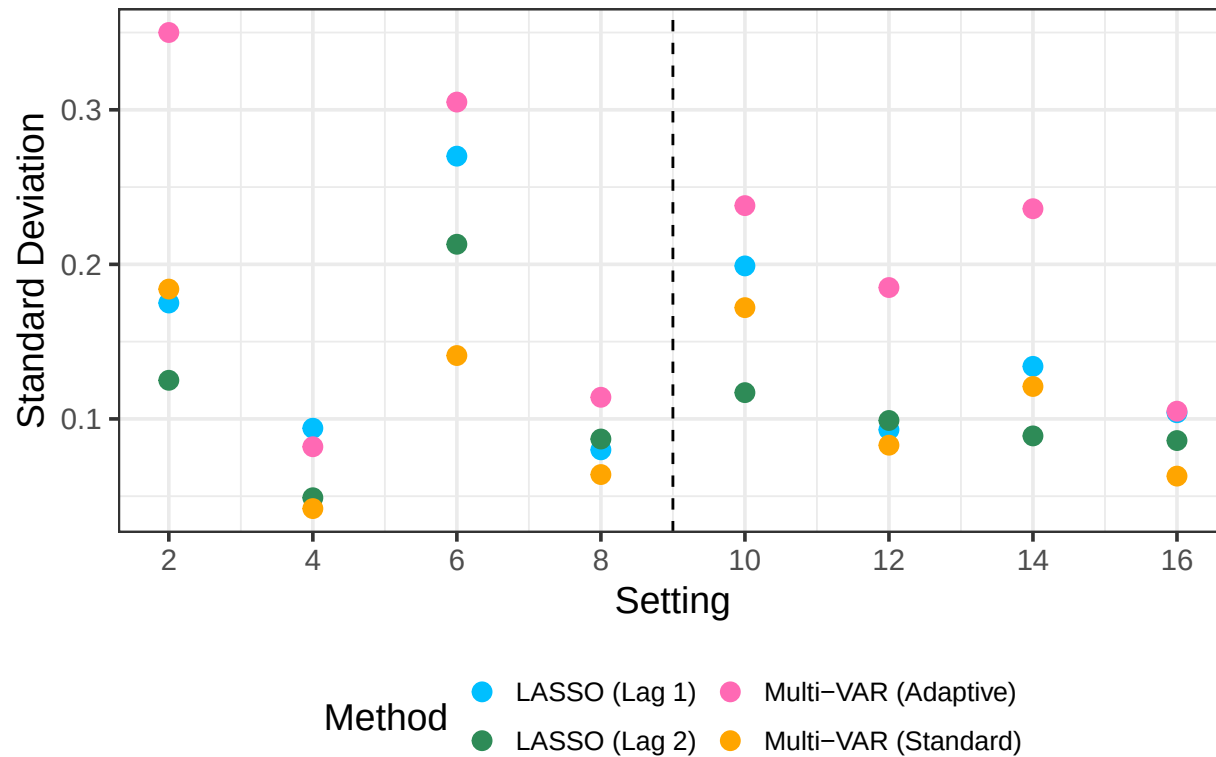
Method

- LASSO (Lag 1)
- Multi-VAR (Standard)
- Multi-VAR (Adaptive)

Means of Sensitivity ($P = 2$)

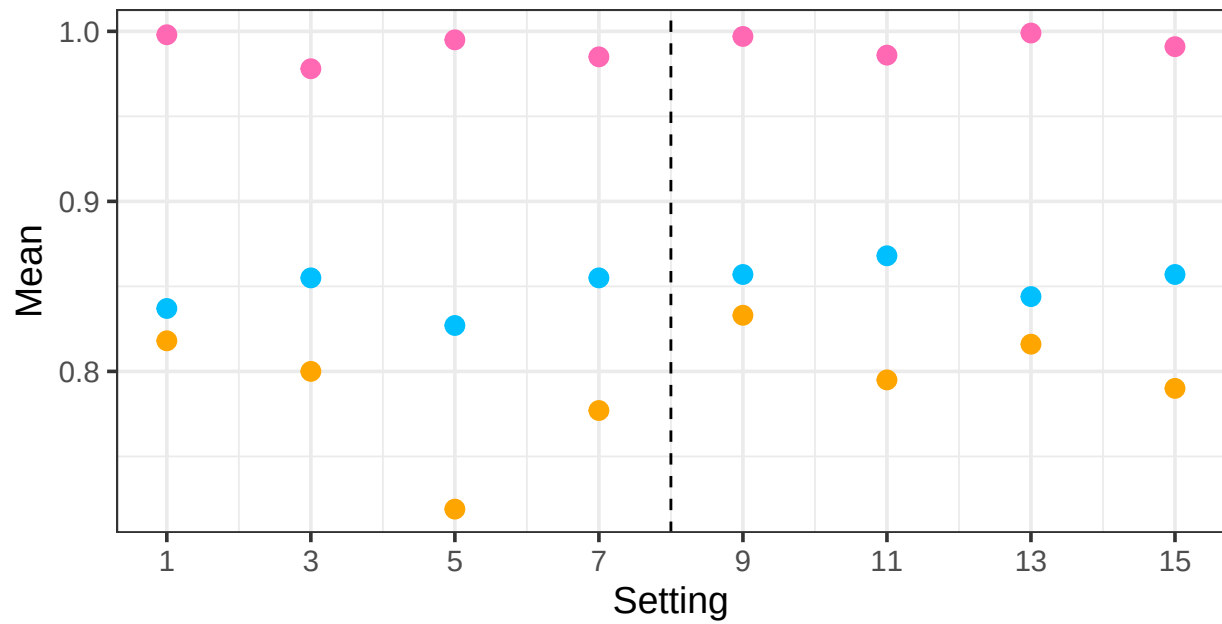


Standard Deviations of Sensitivity ($P = 2$)



SPECIFICITY

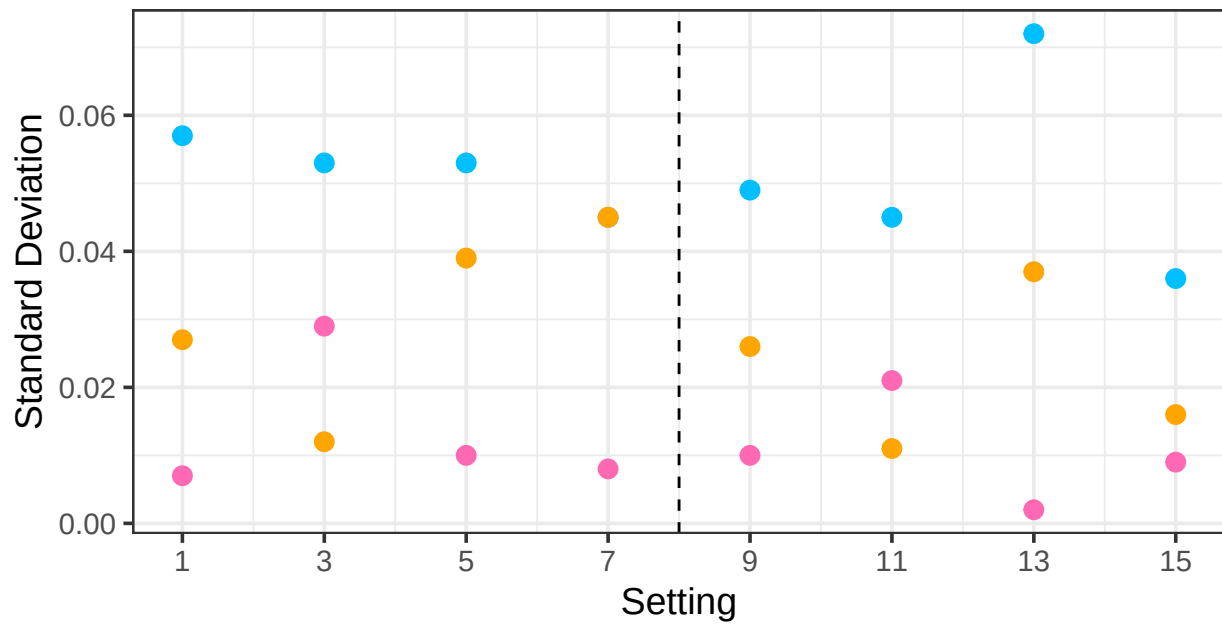
Means of Specificity ($P = 1$)



Method

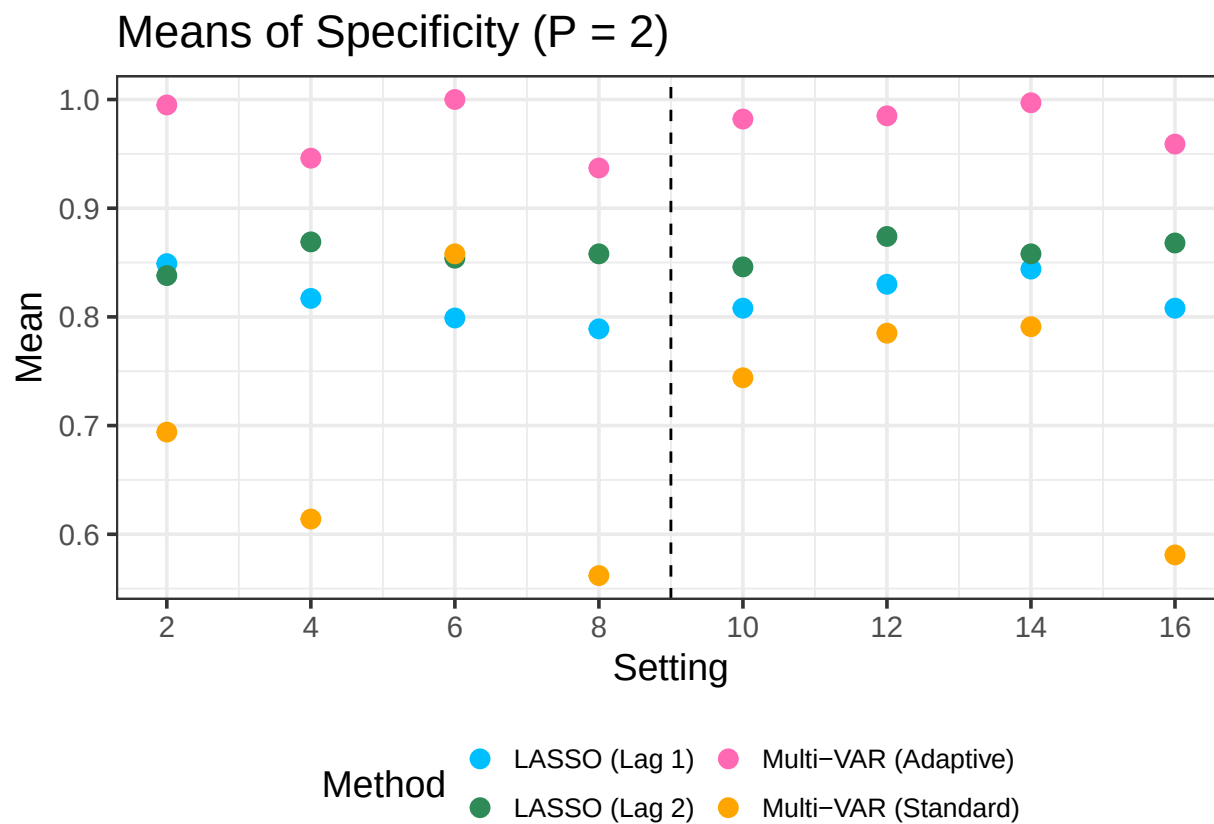
- LASSO (Lag 1)
- Multi-VAR (Standard)
- Multi-VAR (Adaptive)

Standard Deviations of Specificity ($P = 1$)

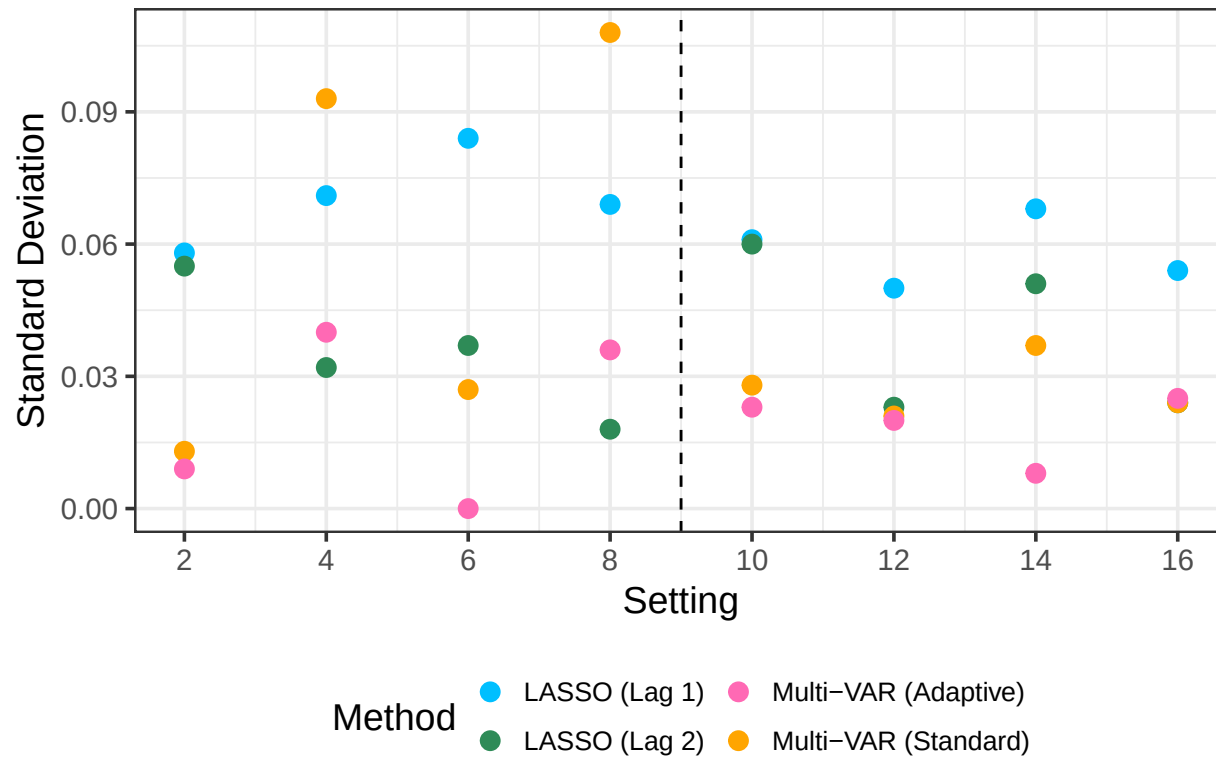


Method

- LASSO (Lag 1)
- Multi-VAR (Standard)
- Multi-VAR (Adaptive)

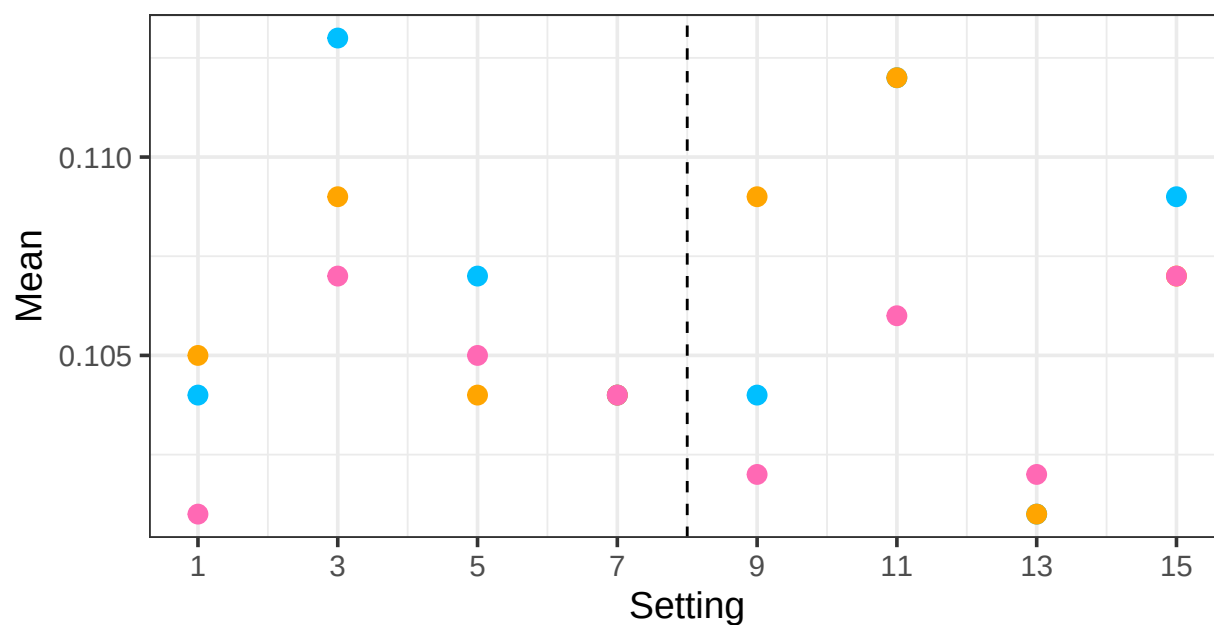


Standard Deviations of Specificity ($P = 2$)



RMSFE

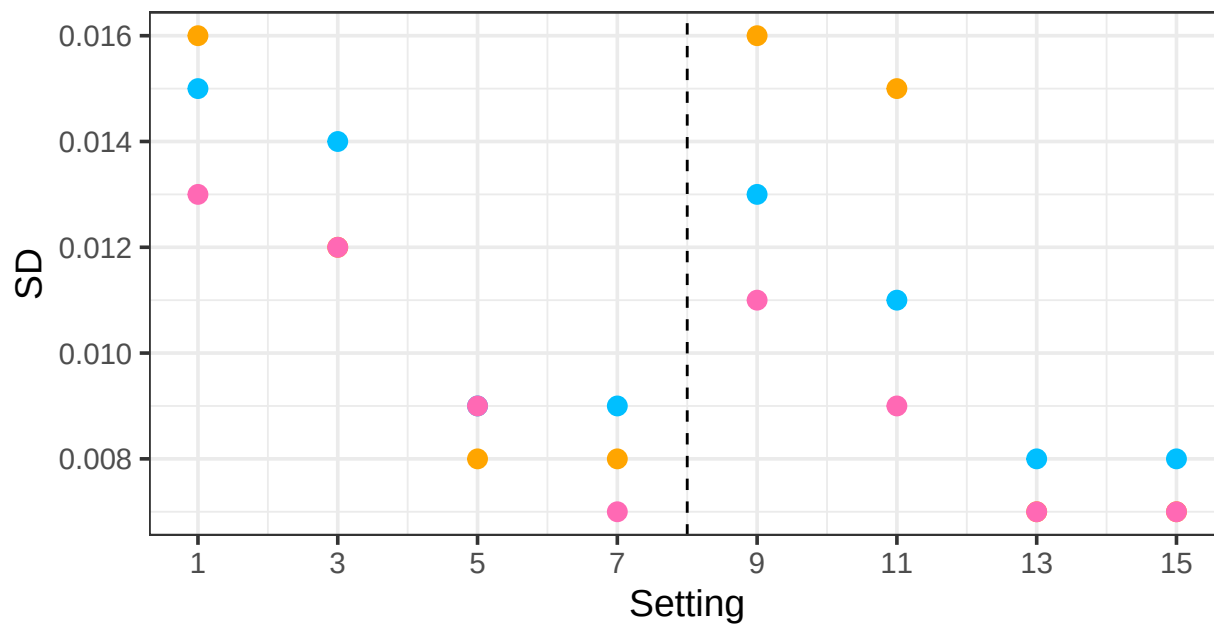
Means of RMSFE (P = 1)



Method

- LASSO (Lag 1)
- Multi-VAR (Standard)
- Multi-VAR (Adaptive)

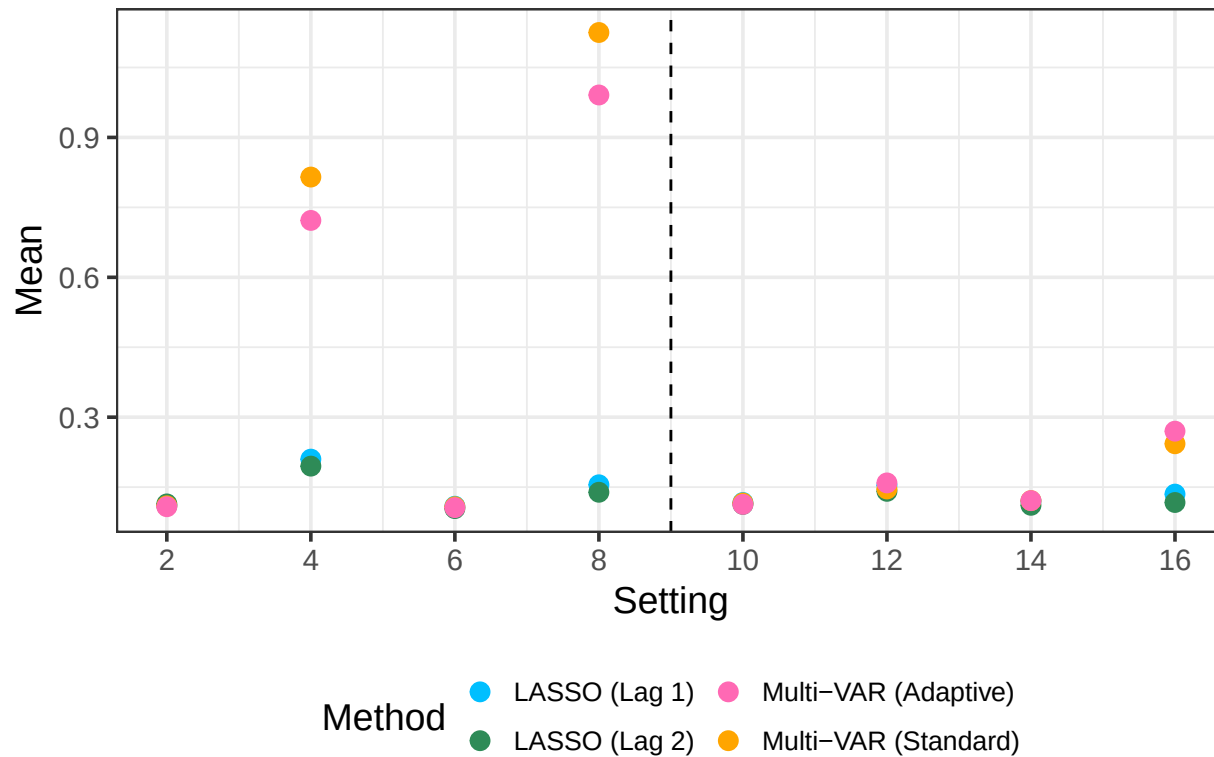
Standard Deviations of RMSFE (P = 1)



Method

- LASSO (Lag 1)
- Multi-VAR (Standard)
- Multi-VAR (Adaptive)

Means of RMSFE (P = 2)



Standard Deviations of RMSFE ($P = 2$)

